Untitled

- PCA
  - ejemplo 1
  - Ejercicio 1 - Error en componentes
  - Ejercicio 2 - Football


```
library(plotly)
library(ggplot2)
library(stringr)
```

PCA

```
n_observaciones <- 100
set.seed(123)
ejex = rnorm(n_observaciones, mean = 10, sd = 2)
ejey = ejex*2.4 + rnorm(length(ejex),mean = 0,sd=1)
ejez = ejex*(-2)*rnorm(n_observaciones, mean = 20, sd = 3) - 0.5 * rnorm(n_observaciones)

ejexx = rnorm(n_observaciones, mean = 10, sd = 2)
ejeyy = ejex*(-4) + rnorm(length(ejex),mean = 0,sd=1)
ejezz = ejex*(4)*rnorm(n_observaciones, mean = 20, sd = 3) - 0.5 * rnorm(n_observaciones)


datos_3d <- data.frame(
  Variable_X = c(ejex,ejexx),
  Variable_Y = c(ejey,ejeyy),
  Variable_Z = c(ejez,ejezz)
)
head(datos_3d)
```

```
##   Variable_X Variable_Y Variable_Z
## 1   8.879049   20.59931  -471.9444
## 2   9.539645   23.15203  -456.3292
## 3  13.117417   31.23511  -503.3593
## 4  10.141017   23.99090  -438.1657
## 5  10.258575   23.66896  -384.6212
## 6  13.430130   32.18728  -498.9944
```

```
cor(datos_3d)
```

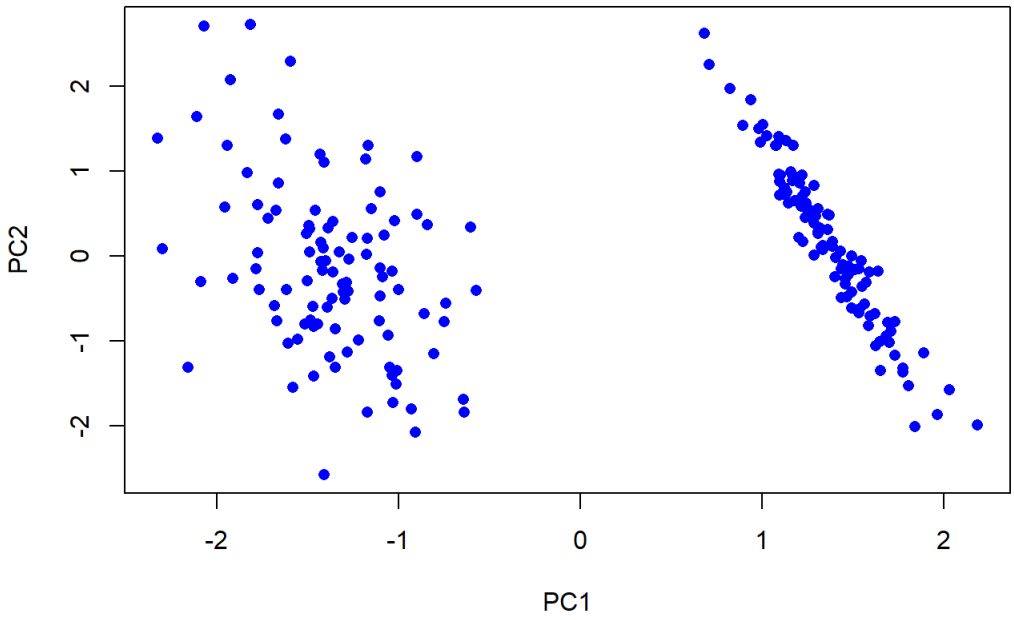
```
##             Variable_X  Variable_Y  Variable_Z
## Variable_X  1.00000000  0.07943584 -0.06408393
## Variable_Y  0.07943584  1.00000000 -0.98696599
## Variable_Z -0.06408393 -0.98696599  1.00000000
```

```
fig <- plot_ly() %>%
  add_markers(
    x = datos_3d[, "Variable_X"],
    y = datos_3d[, "Variable_Y"],
    z = datos_3d[, "Variable_Z"],
    name = "Datos Originales (Escalados)",
    marker = list(size = 4, color = "blue", opacity = 0.8)
  )
fig
```

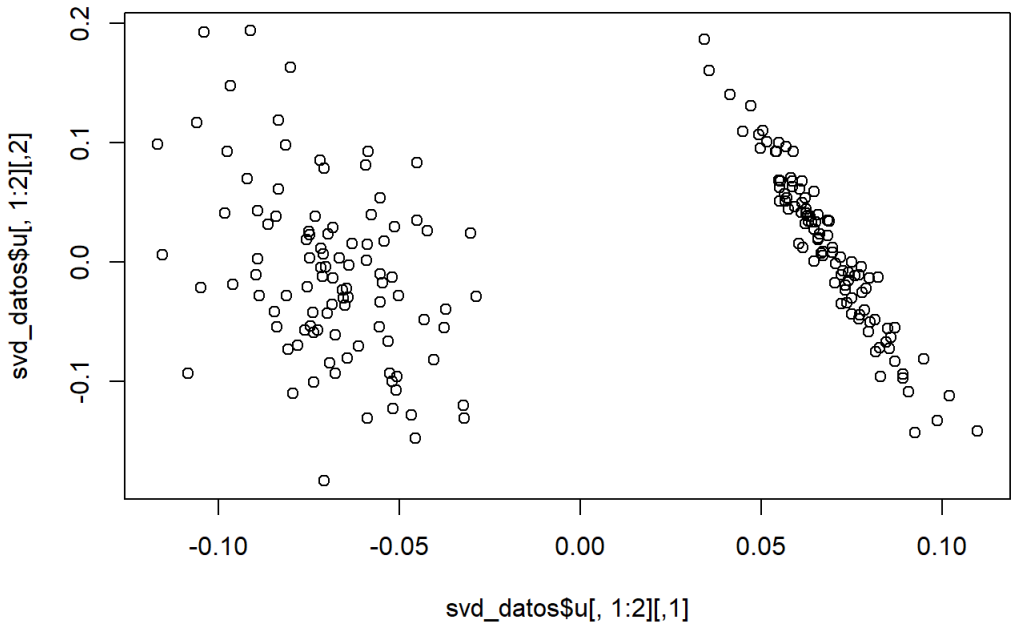
```r
#aplicamos PCA:
pca_resultado <- prcomp(datos_3d, scale = TRUE)
datos_escalados <- scale(datos_3d)
rotaciones <- pca_resultado$rotation



# Longitud del vector para visualización
escala_vector <- 3

# Preparar un data frame con los puntos de inicio (0,0,0) y fin de los 3 vectores PCA
vectores_pca <- data.frame(
  x_inicio = rep(0, 3),
  y_inicio = rep(0, 3),
  z_inicio = rep(0, 3),
  x_fin = rotaciones[1, 1:3] * escala_vector,
  y_fin = rotaciones[2, 1:3] * escala_vector,
  z_fin = rotaciones[3, 1:3] * escala_vector,
  PC = paste0("PC", 1:3)
)

#repetimos la figura con los datos escalados:
fig <- plot_ly() %>%
  add_markers(
    x = datos_escalados[, "Variable_X"],
    y = datos_escalados[, "Variable_Y"],
    z = datos_escalados[, "Variable_Z"],
    name = "Datos Originales (Escalados)",
    marker = list(size = 4, color = "blue", opacity = 0.8)
  )
fig
```

```r
#añadir direccion componentes principales:
colores_pca <- c("PC1" = "red", "PC2" = "green", "PC3" = "orange")
for (i in 1:3) {
  pc_name <- vectores_pca$PC[i]
  pc_color <- colores_pca[pc_name]

  # Añadir la línea del vector (traza)
  fig <- fig %>%
    add_trace(
      x = c(vectores_pca$x_inicio[i], vectores_pca$x_fin[i]),
      y = c(vectores_pca$y_inicio[i], vectores_pca$y_fin[i]),
      z = c(vectores_pca$z_inicio[i], vectores_pca$z_fin[i]),
      type = 'scatter3d',
      mode = 'lines',
      line = list(color = pc_color, width = 5),
      name = pc_name,
      showlegend = TRUE
    ) %>%
    # Añadir la etiqueta al final de la línea (para simular la punta de la flecha)
    add_trace(
      x = vectores_pca$x_fin[i] * 1.1,
      y = vectores_pca$y_fin[i] * 1.1,
      z = vectores_pca$z_fin[i] * 1.1,
      type = 'scatter3d',
      mode = 'text',
      text = pc_name,
      textfont = list(color = pc_color, size = 12),
      name = pc_name,
      showlegend = FALSE
    )
}
fig
```

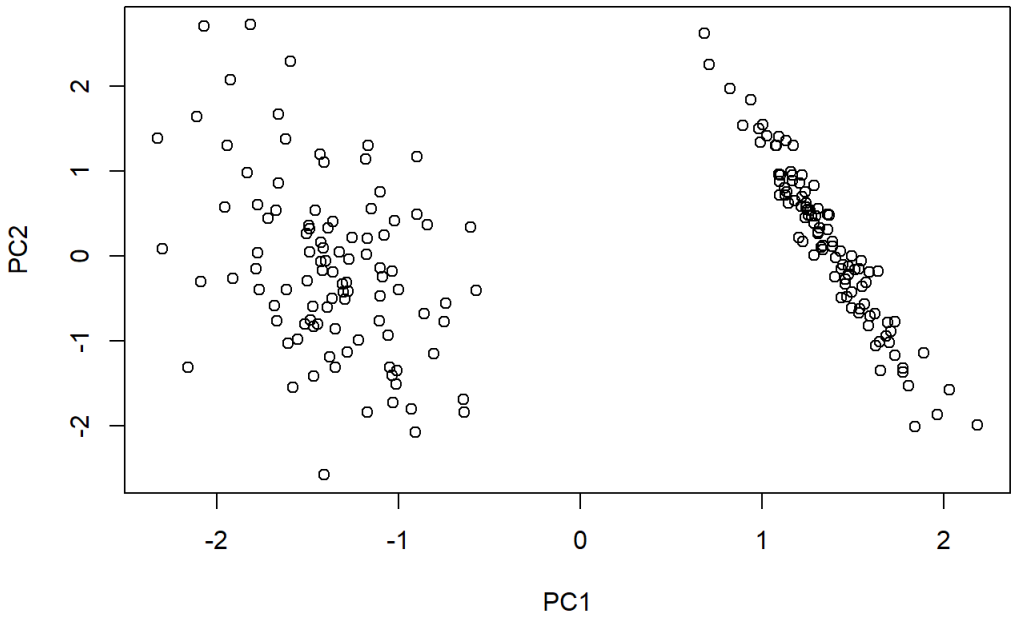
```r
plot(pca_resultado$x[,1:2],pch=16,col="blue")
```




```
#comparacion con funcion svd
pca_resultado <- prcomp(datos_3d, scale = TRUE)
pca_resultado_2 <- prcomp(datos_escalados, scale = FALSE)
svd_datos = svd(datos_escalados)

plot(svd_datos$u[,1:2]) #son equivalentes
```




```
plot(pca_resultado$x[,1:2]) #son equivalente
```



ejemplo 1

```
# Generate random values for x and y
alpha=3
x <- seq(from = 1, to = 5, by = 0.1)
y <- alpha*x
n <- length (x)
x1 <- x + runif(n,-0.25,0.25)
y1<- y + runif(n,-0.25,0.25)
A <- matrix (c(x1,y1), nrow=2, ncol=n, byrow=TRUE)
m <- 2
dim(A) #2 x 41
```

```
## [1]  2 41
```

```
# Calculate princial components with function prcomp
PCs <- prcomp(t(A), center = T, retx = T)
# Eigenvectors
PCs$rotation
```

```
##            PC1        PC2
## [1,] 0.3180516  0.9480734
## [2,] 0.9480734 -0.3180516
```

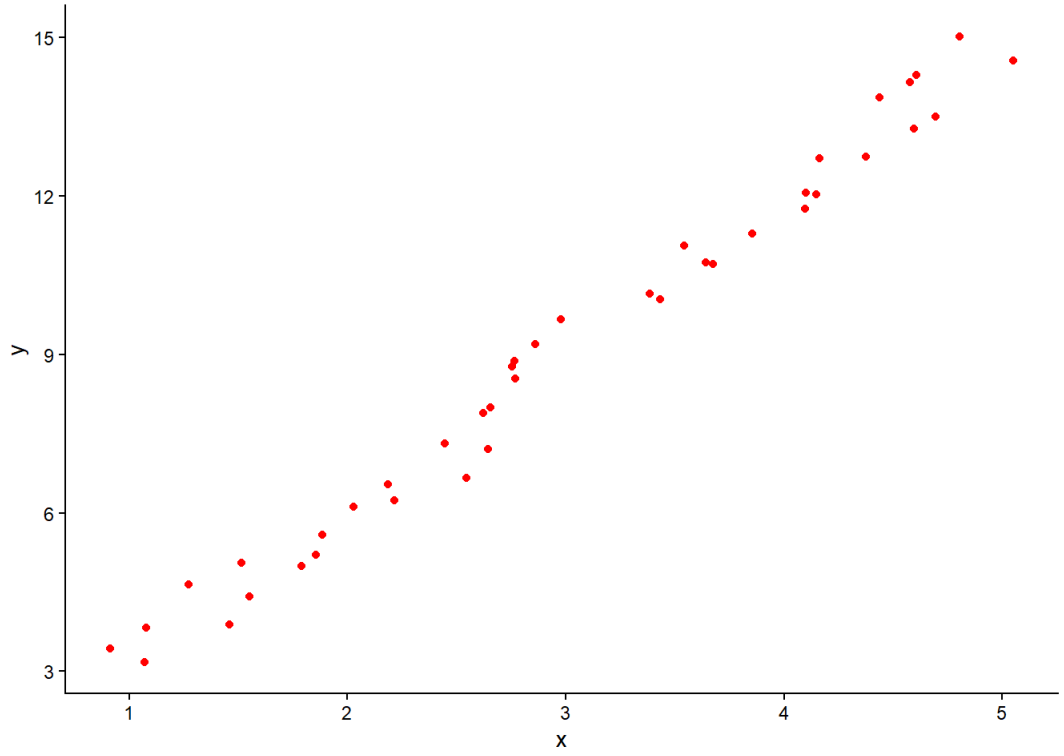
```
#Eigenvalues
PCs$sdev
```

```
## [1] 3.7606410 0.1472074
```

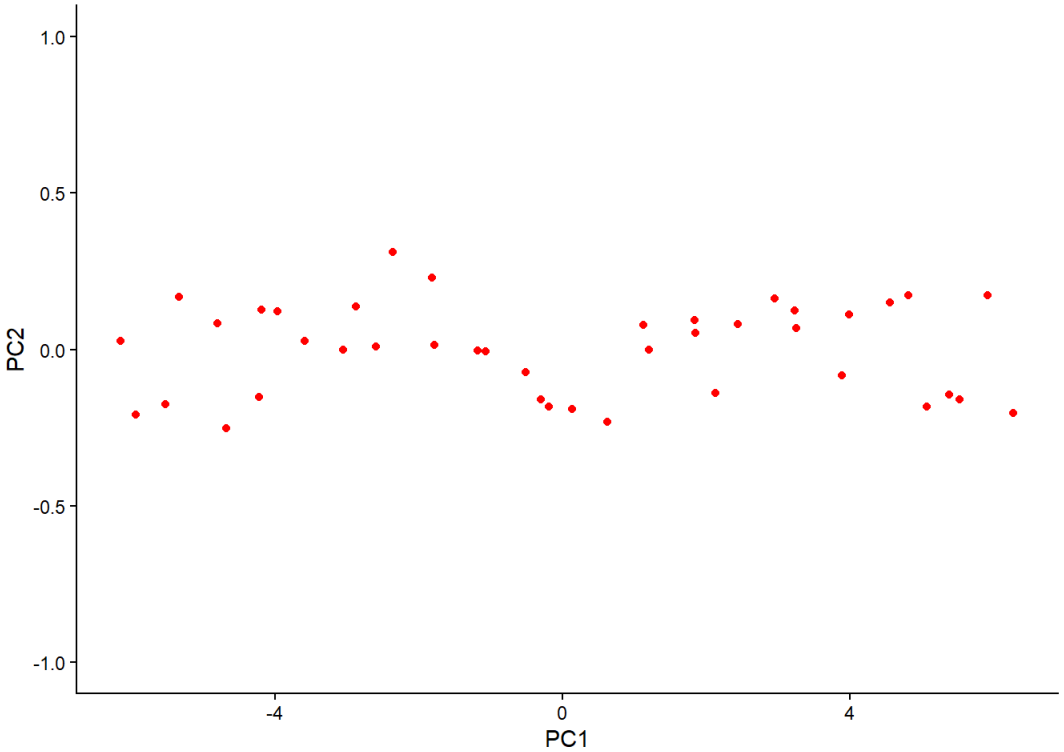
```
# Rotated data (projections)
PCs$x
```

```
##               PC1           PC2
##  [1,] -6.1516807  0.0266522614
##  [2,] -5.9447567 -0.2085554740
##  [3,] -5.5259465 -0.1747833588
##  [4,] -5.3458090  0.1683983321
##  [5,] -4.8080815  0.0846261025
##  [6,] -4.6857078 -0.2506791929
##  [7,] -4.1951467  0.1277986810
##  [8,] -4.2254014 -0.1508071445
##  [9,] -3.9724397  0.1233871462
## [10,] -3.5943304  0.0285732447
## [11,] -3.0530829 -0.0012583132
## [12,] -2.8827861  0.1383803332
## [13,] -2.5954728  0.0100701299
## [14,] -2.3630160  0.3111464080
## [15,] -1.8153857  0.2308303700
## [16,] -1.7829926  0.0135533930
## [17,] -1.0702711 -0.0063828232
## [18,] -1.1801502 -0.0029384522
## [19,] -0.5129839 -0.0729529352
## [20,] -0.1966937 -0.1832681002
## [21,] -0.3032051 -0.1596979498
## [22,]  0.1286578 -0.1897086542
## [23,]  0.6243687 -0.2321324219
## [24,]  1.1206302  0.0791006897
## [25,]  1.2073487 -0.0003728702
## [26,]  1.8380132  0.0948835624
## [27,]  1.8531963  0.0533411537
## [28,]  2.1224881 -0.1397496194
## [29,]  2.4406537  0.0816396387
## [30,]  2.9563057  0.1640556834
## [31,]  3.2316926  0.1249263047
## [32,]  3.2558141  0.0692491216
## [33,]  3.8934951 -0.0819498312
## [34,]  3.9911050  0.1127940820
## [35,]  4.5577625  0.1512428778
## [36,]  4.8111106  0.1739850380
## [37,]  5.0664813 -0.1825633279
## [38,]  5.3877647 -0.1431452333
## [39,]  5.5233853 -0.1591422260
## [40,]  5.9193836  0.1740728531
## [41,]  6.2756835 -0.2026194791
```

```
# Plot original data
data1 <- data.frame(x1, y1)
Fig1 <- ggplot(data1) + geom_point(aes(x=x1, y=y1), color= 'red') +
  xlab('x') + ylab ('y') + theme_classic()
Fig1
```




```
# Plot rotated data
data2 <- data.frame(PC1 = PCs$x[,1], PC2 = PCs$x[,2])
Fig2 <- ggplot(data2) + geom_point(aes(x=PC1, y=PC2), color= 'red') +
  xlab('PC1') + ylab ('PC2') + scale_y_continuous(limits = c(-1,1))+
  theme_classic()
Fig2
```



Ejercicio 1 - Error en componentes

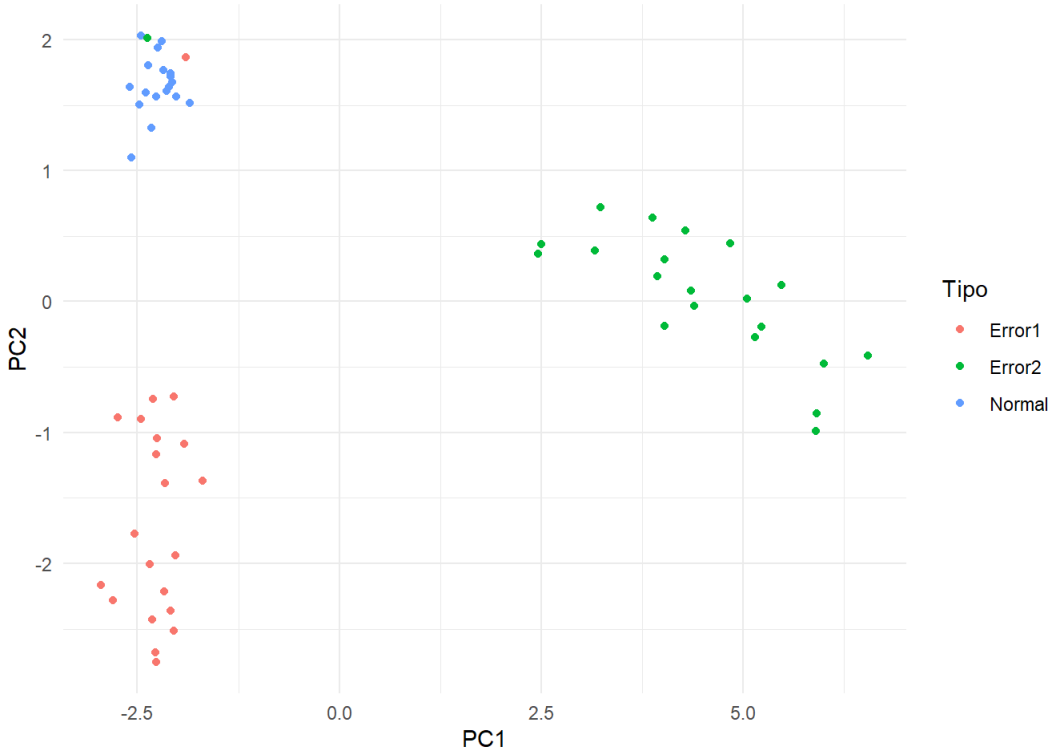
```r
data <- read.csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Juan/Día 3/data/componentes.csv
", header = T, sep = ",")

X <- data[1:14]
tipo <- data$type

PCA <- prcomp(X, scale = T)
summary(PCA)
```

```
## Importance of components:
##                           PC1    PC2     PC3     PC4     PC5     PC6     PC7
## Standard deviation     3.2935 1.4850 0.43511 0.41007 0.36448 0.29973 0.27661
## Proportion of Variance 0.7748 0.1575 0.01352 0.01201 0.00949 0.00642 0.00547
## Cumulative Proportion  0.7748 0.9323 0.94585 0.95786 0.96735 0.97376 0.97923
##                            PC8     PC9    PC10    PC11    PC12    PC13    PC14
## Standard deviation     0.26133 0.23144 0.21388 0.20119 0.19140 0.16192 0.14100
## Proportion of Variance 0.00488 0.00383 0.00327 0.00289 0.00262 0.00187 0.00142
## Cumulative Proportion  0.98411 0.98793 0.99120 0.99409 0.99671 0.99858 1.00000
```

```r
cat("Como podemos comprobar, PC1 tiene una energía de 0.7748 y PC2 de 0.1575 lo que contribuyen en total
un 0.9323 de la energía total")
```

```
## Como podemos comprobar, PC1 tiene una energía de 0.7748 y PC2 de 0.1575 lo que contribuyen en total un
0.9323 de la energía total
```

```r
pc1 <- PCA$x[,1]
pc2 <- PCA$x[,2]

datos_PCA <- data.frame(PC1 = pc1, PC2 = pc2, Tipo = tipo)

ggplot(datos_PCA, aes(x = PC1, y = PC2, color = Tipo)) +
  geom_point() +
  theme_minimal()
```



Ejercicio 2 - Football

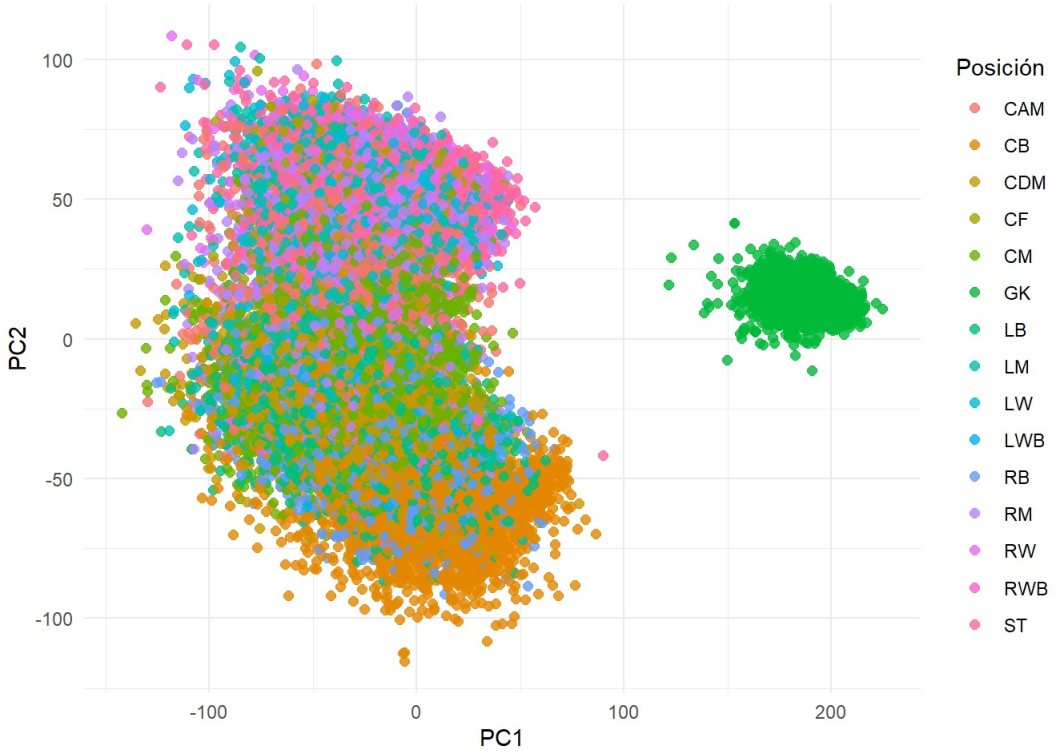
```r
data <- read.csv("C:/Users/enriq/Desktop/Master/Data Science (MIB)/Clases Juan/Día 3/data/FIFA (1).txt",
header = T, sep = ",")

data1 <- data %>%
  select(Acceleration:Volleys, Preferred.Positions) %>%
  mutate(across(Acceleration:Volleys, as.numeric)) %>%
  na.omit()
```

```
## Warning: There were 34 warnings in `mutate()`.
## The first warning was:
## ℹ In argument: `across(Acceleration:Volleys, as.numeric)`.
## Caused by warning:
## ! NAs introducidos por coerción
## ℹ Run `dplyr::last_dplyr_warnings()` to see the 33 remaining warnings.
```

```r
Position <- str_split(data1$Preferred.Positions, " ") %>%
  sapply(function(x) x[1])
data1$Preferred.Positions <- Position



Estadísticas <- select(data1, Acceleration:Volleys)
Posición <- Position

PCA <- prcomp(Estadísticas, scale = F)
summary(PCA)
```

```
## Importance of components:
##                            PC1     PC2      PC3      PC4      PC5      PC6
## Standard deviation     74.8371 43.5787 23.28767 20.58146 16.12477 10.71539
## Proportion of Variance  0.5647  0.1915  0.05468  0.04271  0.02621  0.01158
## Cumulative Proportion   0.5647  0.7561  0.81081  0.85352  0.87973  0.89131
##                             PC7     PC8     PC9   PC10    PC11    PC12    PC13
## Standard deviation     10.17785 9.11852 8.98065 8.5082 8.41550 7.93741 7.15935
## Proportion of Variance  0.01044 0.00838 0.00813 0.0073 0.00714 0.00635 0.00517
## Cumulative Proportion   0.90175 0.91013 0.91827 0.9256 0.93270 0.93906 0.94422
##                           PC14    PC15    PC16    PC17    PC18    PC19    PC20
## Standard deviation     7.06502 6.68497 6.56406 6.50459 6.22369 6.08812 6.00578
## Proportion of Variance 0.00503 0.00451 0.00434 0.00427 0.00391 0.00374 0.00364
## Cumulative Proportion  0.94926 0.95376 0.95811 0.96237 0.96628 0.97001 0.97365
##                           PC21    PC22    PC23    PC24    PC25    PC26   PC27
## Standard deviation     5.91320 5.66946 5.45018 5.15051 4.86761 4.34786 4.1098
## Proportion of Variance 0.00353 0.00324 0.00299 0.00267 0.00239 0.00191 0.0017
## Cumulative Proportion  0.97718 0.98042 0.98341 0.98609 0.98848 0.99038 0.9921
##                           PC28    PC29    PC30    PC31   PC32    PC33    PC34
## Standard deviation     4.05716 3.46035 3.37936 3.31179 3.1429 3.01667 2.95098
## Proportion of Variance 0.00166 0.00121 0.00115 0.00111 0.0010 0.00092 0.00088
## Cumulative Proportion  0.99374 0.99495 0.99610 0.99721 0.9982 0.99912 1.00000
```

```r
pc1 <- PCA$x[,1]
pc2 <- PCA$x[,2]

datos_PCA <- data.frame(PC1 = pc1, PC2 = pc2, Posición = Posición)

ggplot(datos_PCA, aes(x = PC1, y = PC2, color = Posición)) +
  geom_point(size = 2, alpha = 0.8) +
  theme_minimal()
```


Posición
CAM
CB
CDM
CF
CM
GK
LB
LM
LW
LWB
RB
RM
RW
RWB
ST
PC2
PC1
100
50
0
-50
-100
-100
0
100
200